PROJECT TOPIC: AI Study Planner

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Assignment-06: Prepare Implementation Plan for Mini Project

1. Project Objective

The objective of this project is to design and implement an AI Study Planner, a web-based application that helps a student prepare for examinations by generating a study schedule that is based on the student's own subjects, topics, exam dates, available study hours and topic difficulty. Instead of a static timetable that stays the same throughout the exam preparation period, this system is meant to update the schedule whenever new information about the student's performance becomes available, so that weaker topics get more attention and topics that are already understood get comparatively less time. The plan below explains how this idea will actually be built, step by step.

2. System Architecture / Overall Workflow

The system works in a simple cycle rather than a one-time process. A student first enters basic information about the subjects and topics to be studied. Based on this information along with the exam dates and available hours, the system calculates a priority for every topic and generates an initial study schedule. The student then studies according to this schedule and, after completing a topic, attempts a short quiz. The result of this quiz is used to update how well the student actually understands that topic, and the schedule is recalculated for the remaining days. This overall flow can be summarised as follows:

Student Information → Subjects and Topics → Exam Dates and Available Study Hours → Topic Priority Calculation → Study Schedule → Student Studies → Quiz/Test → Performance Analysis → Updated Topic Priority → Adaptive Study Schedule

At a technical level, the system will have three main parts. The frontend is the interface where the student enters details and views the schedule. The backend handles the logic of calculating priority and generating the schedule. The database stores the student's subjects, topics, quiz scores and the schedule itself so that it can be updated instead of recreated from scratch every time.

3. Input Module

The input module is the starting point of the system. Through this module, the student provides the following details:

- Name and basic profile information of the student
- List of subjects being studied, along with the topics under each

subject • Exam date for each subject

- Difficulty level of each topic, rated by the student on a simple scale such as easy, medium or hard
- Number of hours the student can realistically devote to study each day or each

week These inputs form the base on which the rest of the system depends, so the input module also needs basic validation, such as not allowing an exam date in the past or an available study time of zero hours for a subject that still needs to be completed.

4. Data Processing

Once the student has entered the required information, the system processes it before any calculation is done. This involves calculating the number of days remaining until each exam, converting the difficulty rating entered by the student into a numeric value that can be used in a formula, and organising the topics subject-wise so that the available study hours can later be distributed properly. This processed data is stored in the database and is what the priority calculation module works on in the next step.

5. Topic Priority Calculation

The core of the system is the calculation of a priority score for every topic. In this project, the priority score will be calculated using a rule-based formula rather than a complex machine learning model, since this is more realistic to implement and evaluate within the scope of a college mini project. A simple conceptual formula that will be used is:

Priority Score = (Weight1 × Exam Urgency) + (Weight2 × Topic Difficulty) + (Weight3 × Student Weakness)

Exam urgency increases as the exam date comes closer, topic difficulty is taken from what the student has entered, and student weakness is derived from quiz performance, being higher for topics the student has performed poorly in. The weights (Weight1, Weight2, Weight3) decide how much importance each factor is given and can be adjusted while testing the system. A machine learning based approach, where the system learns these weights automatically from a student's past data, is mentioned here only as a possible future enhancement and is not something this mini project claims to fully implement.

6. Study Schedule Generation

Once priority scores are calculated for all topics, the system distributes the student's available study hours among the topics in proportion to their priority. Topics with a higher score are given more time in the schedule, while topics with a lower score are given comparatively less time, but every topic still receives at least a minimum amount of time so that nothing is completely left out. The schedule is generated day-wise for the period leading up to each exam, and it is designed to stay within the number of hours the student has stated they can realistically study.

7. Performance Assessment / Quiz Module

After a student has studied a topic according to the schedule, the system provides a short quiz for that topic, made up of a small set of objective or short-answer questions stored in the database. The score obtained in this quiz is recorded and treated as an indicator of how well the student currently understands the topic. This is one of the more important parts of the system, since it is the main source of feedback that makes the schedule adaptive instead of fixed.

8. Adaptive Rescheduling

Whenever a new quiz score is recorded, the weakness value for that topic is updated. A low score increases the weakness value and therefore increases the priority score of that topic, while a high score reduces it. The system then recalculates the study schedule for the remaining days before the exam, so that topics the student is struggling with automatically receive more time in the upcoming days, without the student having to manually edit the timetable.

9. Technology Stack

The project is planned to be built using commonly available and well-documented technologies so that it can be completed within the time available for a mini project.

- Frontend: HTML, CSS, JavaScript, or a simple React interface
- Backend: Node.js with Express, or Python with Flask
- Database: MySQL or Firebase, for storing student data, topics and quiz results • Future scope: a basic machine learning model such as linear regression, to predict weakness more accurately from historical quiz data

10. Implementation Phases

The implementation of this project will be carried out in the following phases: • Phase 1: Requirement gathering and designing the basic user interface • Phase 2: Building the input module and setting up the database structure • Phase 3:

Implementing the priority calculation formula and the schedule generation logic •

Phase 4: Building the quiz module and connecting it to the topics

- Phase 5: Implementing adaptive rescheduling based on quiz results
- Phase 6: Testing the complete system and fixing issues found during testing

11. Testing and Evaluation

Testing will be done at two levels. At the module level, the priority formula and the schedule generation logic will be tested using sample subjects and topics to check that the calculated hours do not exceed the available study hours entered by the student, and that topics with higher priority actually receive more time. At the system level, the group members will use their own subjects and exam dates as sample data to check whether the generated schedule looks reasonable and whether the schedule updates correctly after a quiz is attempted.

12. Expected Outcome

By the end of this project, we expect to have a working prototype of the AI Study Planner that can take a student's subjects, topics, exam dates and available hours as input, generate an initial study schedule, and update this schedule based on quiz performance. The expected outcome is a functioning demonstration of the idea described in the problem statement, not a claim that using this tool will guarantee better exam results. The system is meant to reduce the manual effort of planning and revising a study schedule and to make the distribution of study time more consistent with exam urgency, topic difficulty and actual performance.